

Lab 06 – Intel-Driven Hunt in the Bachelor Lab

Industrial Security (Master)

• Maps to Section 06

• 2–3 hours

LAB 06

🎯 Goal

Take a real, recent OT-relevant ATT&CK technique. Form a hunt hypothesis. Execute it against the Bachelor Docker lab Zeek logs. Iterate until you can either confirm or deny.

✂ Step 1 – Pick a TTP

From the ATT&CK for ICS matrix, pick one of: T0855 (Unauthorised Command Message), T0859 (Valid Accounts), T0846 (Remote System Discovery), T0867 (Lateral Tool Transfer). Note the canonical detection guidance.

✂ Step 2 – Translate to lab telemetry

For your chosen TTP, identify what would be visible in: Zeek `conn.log`, `modbus.log`, `notice.log`, or in OpenPLC's web-UI access log.

✂ Step 3 – Generate baseline

```
1 cd bachelor/labs/_stack && docker compose up -d
2 docker compose exec student python3 /labs/scripts/modbus_poll_loop.py &
3 sleep 300
```

✂ Step 4 – Inject and hunt

Trigger your chosen TTP (e.g. `modbus_inject.py`). Search the Zeek logs for the signal. Document the time-to-detection and the precision (false positives in 5 min of baseline).

✂ Step 5 – Refine

If your hunt was noisy, tighten the query. If it missed, broaden it. Document the iteration.

📁 Deliverable

A 1-page hunt report: TTP, hypothesis, query, baseline-noise level, detection result, refinement. Bonus: a Zeek script that turns the hunt into a continuous notice.